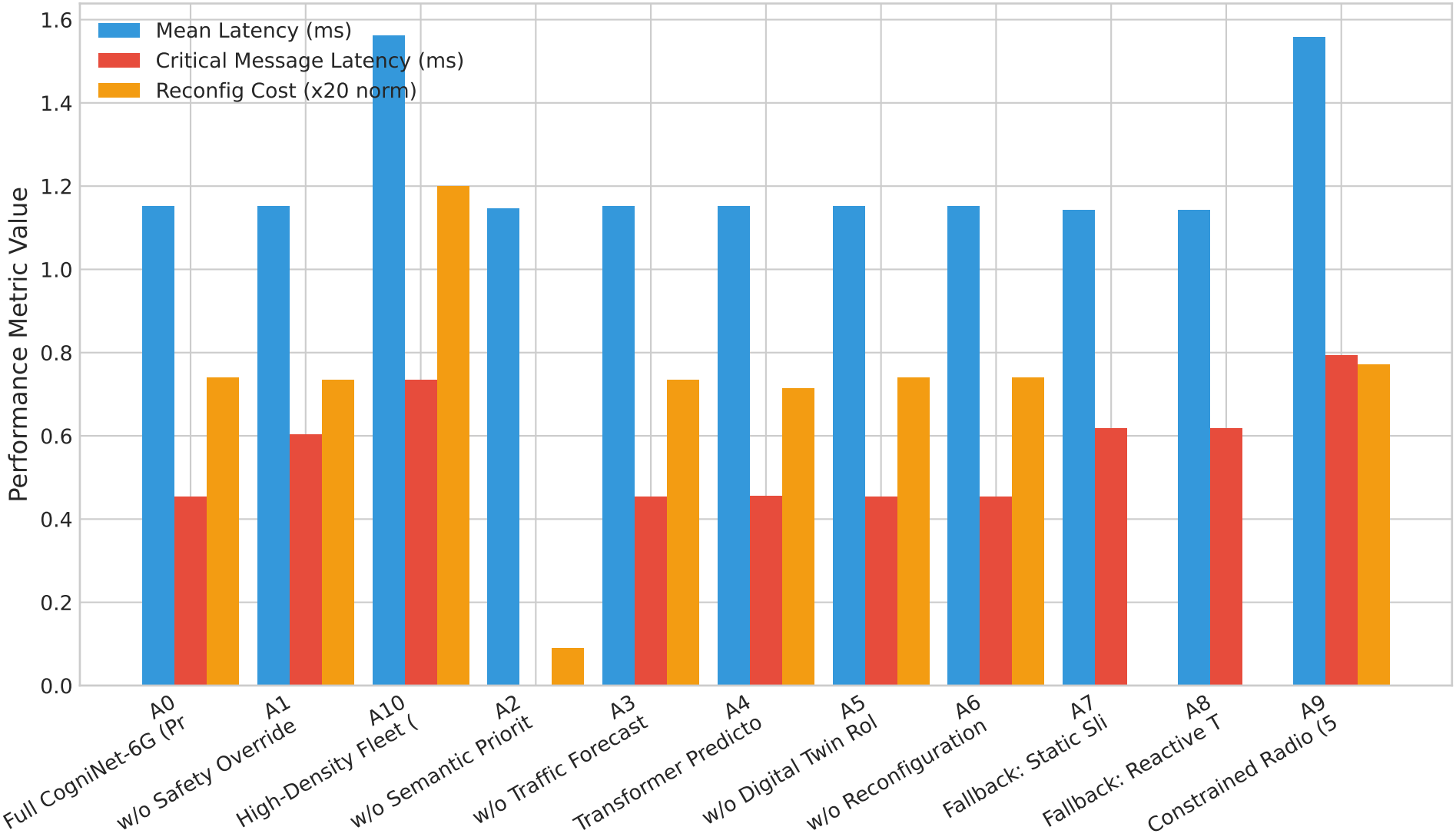
Ablation Study: Component Impact on System Latency & Reconfiguration Cost
Mean Latency (ms)
Critical Message Latency (ms)
Reconfig Cost (x20 norm)
Performance Metric Value
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
A0
Full CogniNet-6G (Pr
A1
w/o Safety Override
A10
High-Density Fleet (
A2
w/o Semantic Priorit
A3
w/o Traffic Forecast
A4
Transformer Predicto
A5
w/o Digital Twin RoI
A6
w/o Reconfiguration
A7
Fallback: Static Sli
A8
Fallback: Reactive T
A9
Constrained Radio (5